Limit: Questions and Full Solutions

Question 01

English

Let P and Q be two points on the circle $x^2 + y^2 = 1$ with centre O such that $\angle POQ = \theta$, where $0 < \theta < \dfrac{\pi}{2}$. The tangents to the circle at P and Q meet at T. Let $A(\theta)$ denote the area of the region bounded by the chord PQ and the minor arc PQ, and let $B(\theta) = PT + TQ - PQ$. Then $\lim\limits_{\theta \to 0^+} \dfrac{A(\theta)}{B(\theta)}$ is equal to

(A) $\dfrac{1}{3}$

(B) $\dfrac{1}{2}$

(C) $\dfrac{2}{3}$

(D) $\dfrac{4}{3}$

Only one option is correct.

हिन्दी

माना कि P तथा Q वृत्त $x^2 + y^2 = 1$ (केन्द्र O) पर दो बिन्दु इस प्रकार हैं कि $\angle POQ = \theta$, जहाँ $0 < \theta < \dfrac{\pi}{2}$ है। P तथा Q पर वृत्त की स्पर्श रेखाएँ T पर मिलती हैं। माना कि $A(\theta)$ जीवा PQ तथा लघु चाप PQ से परिबद्ध क्षेत्र का क्षेत्रफल है, तथा $B(\theta) = PT + TQ - PQ$ है। तब $\lim\limits_{\theta \to 0^+} \dfrac{A(\theta)}{B(\theta)}$ बराबर है

(A) $\dfrac{1}{3}$

(B) $\dfrac{1}{2}$

(C) $\dfrac{2}{3}$

(D) $\dfrac{4}{3}$

केवल एक विकल्प सही है।

Solution

Area of sector OPQ is $\dfrac{\theta}{2}$ (radius 1) and area of triangle OPQ is $\dfrac{1}{2}\sin\theta$. The region between chord PQ and the minor arc is the sector minus the triangle, so $A(\theta) = \dfrac{\theta - \sin\theta}{2}$.

Since $OP \perp PT$, $OQ \perp QT$ and $OP = OQ$, the line OT bisects $\angle POQ$. In right triangle OPT, $\angle POT = \dfrac{\theta}{2}$, so $PT = \tan\dfrac{\theta}{2}$, and similarly $TQ = \tan\dfrac{\theta}{2}$.

The chord is $PQ = 2\sin\dfrac{\theta}{2}$. Hence, with $u = \dfrac{\theta}{2}$, $B(\theta) = 2\tan u - 2\sin u = 2(\tan u - \sin u)$.

Expansions: $\tan u = u + \dfrac{u^3}{3} + \ldots$ and $\sin u = u - \dfrac{u^3}{6} + \ldots$, so $\tan u - \sin u = \dfrac{u^3}{2} + O(u^5)$ and $B(\theta) = u^3 + O(u^5) = \dfrac{\theta^3}{8} + O(\theta^5)$.

Also $\sin\theta = \theta - \dfrac{\theta^3}{6} + \ldots$, so $\theta - \sin\theta = \dfrac{\theta^3}{6} + O(\theta^5)$ and $A(\theta) = \dfrac{\theta^3}{12} + O(\theta^5)$.

Therefore $\dfrac{A(\theta)}{B(\theta)} = \dfrac{\frac{1}{12} + O(\theta^2)}{\frac{1}{8} + O(\theta^2)} \to \dfrac{8}{12} = \dfrac{2}{3}$.

(Taking $B \approx \dfrac{\theta^3}{4}$, i.e. forgetting the half angle, gives $\dfrac{1}{3}$; taking $A \approx \dfrac{\theta^3}{6}$ gives $\dfrac{4}{3}$.)

Correct Option: (C)

Question 02

English

The value of $\lim\limits_{x \to \pi/6} (2\cos 2x - 1)\tan 3x$ is

(A) $\dfrac{2}{\sqrt{3}}$

(B) $\dfrac{\sqrt{3}}{2}$

(C) $-\dfrac{2}{\sqrt{3}}$

(D) 2

Only one option is correct.

हिन्दी

$\lim\limits_{x \to \pi/6} (2\cos 2x - 1)\tan 3x$ **का मान है**

(A) $\dfrac{2}{\sqrt{3}}$

(B) $\dfrac{\sqrt{3}}{2}$

(C) $-\dfrac{2}{\sqrt{3}}$

(D) 2

केवल एक विकल्प सही है।

Solution

At $x = \dfrac{\pi}{6}$, $2\cos 2x - 1 = 2 \cdot \dfrac{1}{2} - 1 = 0$, while $\tan 3x$ is unbounded because $3x \to \dfrac{\pi}{2}$. The limit is of the form $0 \cdot \infty$.

Triple-angle formula: $\cos 3x = 4\cos^3 x - 3\cos x = \cos x\,(4\cos^2 x - 3)$.

Double-angle formula: $4\cos^2 x = 2(1 + \cos 2x)$, so $4\cos^2 x - 3 = 2\cos 2x - 1$. Hence $\cos 3x = \cos x\,(2\cos 2x - 1)$, that is, $2\cos 2x - 1 = \dfrac{\cos 3x}{\cos x}$ (valid because $\cos x \neq 0$ near $\dfrac{\pi}{6}$).

For x near $\dfrac{\pi}{6}$ with $x \neq \dfrac{\pi}{6}$, $\cos 3x \neq 0$ and $(2\cos 2x - 1)\tan 3x = \dfrac{\cos 3x}{\cos x} \cdot \dfrac{\sin 3x}{\cos 3x} = \dfrac{\sin 3x}{\cos x}$.

This is continuous at $\dfrac{\pi}{6}$, so the limit is $\dfrac{\sin(\pi/2)}{\cos(\pi/6)} = \dfrac{1}{\sqrt{3}/2} = \dfrac{2}{\sqrt{3}}$. The cancelled factor $\cos 3x$ changes sign at $\dfrac{\pi}{6}$ in both factors, so the left and right limits agree.

(A sign slip in L'Hopital's rule gives $-\dfrac{2}{\sqrt{3}}$; using $\sin x$ in place of $\cos x$ in the denominator gives 2.)

Correct Option: (A)

Question 03

English

Let $N : \left(-\frac{\pi}{4}, \frac{\pi}{4}\right) \to \mathbb{R}$ be defined by $N(t) = t - \tan t$. Then $\lim\limits_{x\to 0} \frac{N(N(x))}{x^9}$ is equal to

(A) $-\frac{1}{81}$

(B) $\frac{1}{81}$

(C) $\frac{1}{27}$

(D) $-\frac{1}{27}$

Only one option is correct.

हिन्दी

माना कि $N : \left(-\frac{\pi}{4}, \frac{\pi}{4}\right) \to \mathbb{R}$, $N(t) = t - \tan t$ द्वारा परिभाषित है। तब $\lim\limits_{x\to 0} \frac{N(N(x))}{x^9}$ बराबर है

(A) $-\frac{1}{81}$

(B) $\frac{1}{81}$

(C) $\frac{1}{27}$

(D) $-\frac{1}{27}$

केवल एक विकल्प सही है।

Solution

Near 0, $N(x)$ is small, so $N(N(x))$ is defined. Differentiating $N(N(x))$ nine times is impractical, so the Maclaurin series of tan is used.

$\tan t = t + \frac{t^3}{3} + \frac{2t^5}{15} + \ldots$, hence $N(t) = t - \tan t = -\frac{t^3}{3} - \frac{2t^5}{15} - \ldots$ and $\lim\limits_{t\to 0} \frac{N(t)}{t^3} = -\frac{1}{3}$.

Put $u = N(x)$. Then $u \to 0$ as $x \to 0$, and $u \neq 0$ for small $x \neq 0$ (since $\tan x \neq x$ there). From the series, $u = -\frac{x^3}{3}\left(1 + \frac{2x^2}{5} + \ldots\right)$, so $\frac{u^3}{x^9} \to \left(-\frac{1}{3}\right)^3 = -\frac{1}{27}$.

Write $\frac{N(N(x))}{x^9} = \frac{N(u)}{u^3} \cdot \frac{u^3}{x^9}$. The first factor tends to $-\frac{1}{3}$ (because $u \to 0$) and the second to $-\frac{1}{27}$.

Hence the limit is $\left(-\frac{1}{3}\right)\left(-\frac{1}{27}\right) = \frac{1}{81}$. (Dropping one sign gives $-\frac{1}{81}$; treating $N(N(x))$ as $N(x)^3$ gives $-\frac{1}{27}$.)

Correct Option: (B)

Question 04

English

For $n \in \mathbb{N}$, let $a_n = \dfrac{1}{4} \cdot \dfrac{\binom{2n+2}{n+1}}{\binom{2n}{n}}$. Then $\lim\limits_{n\to\infty} (a_n)^n$ is equal to

(A) e^{-1}
(B) $e^{1/2}$
(C) 1
(D) $e^{-1/2}$

Only one option is correct.

हिन्दी

$n \in \mathbb{N}$ के लिए माना कि $a_n = \dfrac{1}{4} \cdot \dfrac{\binom{2n+2}{n+1}}{\binom{2n}{n}}$ है। तब $\lim\limits_{n\to\infty} (a_n)^n$ बराबर है

(A) e^{-1}
(B) $e^{1/2}$
(C) 1
(D) $e^{-1/2}$

केवल एक विकल्प सही है।

Solution

$\binom{2n+2}{n+1} = \dfrac{(2n+2)!}{(n+1)!\,(n+1)!} = \dfrac{(2n+2)(2n+1)}{(n+1)(n+1)} \cdot \dfrac{(2n)!}{n!\,n!}$, so the ratio is $\dfrac{(2n+2)(2n+1)}{(n+1)^2} = \dfrac{2(2n+1)}{n+1}$.

Hence $a_n = \dfrac{1}{4} \cdot \dfrac{2(2n+1)}{n+1} = \dfrac{2n+1}{2n+2} = 1 - \dfrac{1}{2n+2}$, which tends to 1, while the exponent $n \to \infty$: the form is 1^∞.

For $a_n \to 1$ and $b_n \to \infty$, $\lim a_n^{b_n} = e^{\lim b_n(a_n-1)}$ (from $\ln a_n \sim a_n - 1$).

Here $b_n(a_n - 1) = n\left(-\dfrac{1}{2n+2}\right) = -\dfrac{n}{2n+2} \to -\dfrac{1}{2}$.

Therefore $\lim\limits_{n\to\infty} (a_n)^n = e^{-1/2}$. (Answering 1 treats 1^∞ as 1; dropping the factor 2 in $2n+2$ gives e^{-1}.)

Correct Option: (D)

Question 05

English

For $n \in \mathbb{N}$, let $S_n = \sum_{k=1}^{n} \frac{n}{n^2+k}$. Then $\lim_{n\to\infty} n\,(S_n - 1)$ is equal to

(A) $-\frac{1}{2}$
(B) $\frac{1}{2}$
(C) 0
(D) -1

Only one option is correct.

हिन्दी

$n \in \mathbb{N}$ के लिए माना कि $S_n = \sum_{k=1}^{n} \frac{n}{n^2+k}$ है। तब $\lim_{n\to\infty} n\,(S_n - 1)$ बराबर है

(A) $-\frac{1}{2}$
(B) $\frac{1}{2}$
(C) 0
(D) -1

केवल एक विकल्प सही है।

Solution

First, $\frac{n}{n^2+n} \le \frac{n}{n^2+k} \le \frac{n}{n^2+1}$, so $\frac{n^2}{n^2+n} \le S_n \le \frac{n^2}{n^2+1}$ and $S_n \to 1$. This makes $n(S_n - 1)$ a $0 \cdot \infty$ form, so the first-order bound is not enough.

Subtract $\frac{1}{n}$ from each of the n terms: $S_n - 1 = \sum_{k=1}^{n} \left(\frac{n}{n^2+k} - \frac{1}{n} \right) = \sum_{k=1}^{n} \frac{n^2 - (n^2+k)}{n(n^2+k)} = -\sum_{k=1}^{n} \frac{k}{n(n^2+k)}$.

Hence $n(S_n - 1) = -T_n$ with $T_n = \sum_{k=1}^{n} \frac{k}{n^2+k}$.

For $1 \le k \le n$, $n^2+1 \le n^2+k \le n^2+n$, so $\frac{k}{n^2+n} \le \frac{k}{n^2+k} \le \frac{k}{n^2+1}$. Summing with $\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$:

$$\frac{n(n+1)}{2(n^2+n)} \le T_n \le \frac{n(n+1)}{2(n^2+1)}$$

The left bound equals $\frac{1}{2}$. The right bound is a rational function whose numerator and denominator both have leading term n^2, giving $\frac{n^2}{2n^2} \to \frac{1}{2}$.

By the sandwich theorem $T_n \to \frac{1}{2}$, so $\lim_{n\to\infty} n(S_n - 1) = -\frac{1}{2}$. (Stopping at $S_n \to 1$ suggests 0; losing the sign gives $\frac{1}{2}$.)

Correct Option: (A)

Question 06

English

Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(y) = \dfrac{1 - \cos y}{y^2}$ for $y \neq 0$ and $f(0) = 1$, and let $[t]$ denote the greatest integer less than or equal to t. Then $\lim\limits_{x \to 0} \Big(f\big([\cos x]\big) + f(\sin x) \Big)$ is equal to

(A) 1
(B) $\dfrac{3}{2}$
(C) 2
(D) $\dfrac{3}{2} - \cos 1$

Only one option is correct.

हिन्दी

माना कि $f : \mathbb{R} \to \mathbb{R}$, $y \neq 0$ के लिए $f(y) = \dfrac{1 - \cos y}{y^2}$ तथा $f(0) = 1$ द्वारा परिभाषित है, तथा $[t]$, t से कम या उसके बराबर महत्तम पूर्णांक को दर्शाता है। तब $\lim\limits_{x \to 0} \Big(f\big([\cos x]\big) + f(\sin x) \Big)$ बराबर है

(A) 1
(B) $\dfrac{3}{2}$
(C) 2
(D) $\dfrac{3}{2} - \cos 1$

केवल एक विकल्प सही है।

Solution

The two composite limits behave differently, and each must be justified separately.

First term. For $0 < |x| < \dfrac{\pi}{2}$ we have $0 < \cos x < 1$, hence $[\cos x] = 0$ and $f([\cos x]) = f(0) = 1$. The function $x \mapsto f([\cos x])$ is constant $(= 1)$ on a punctured neighbourhood of 0, so its limit is 1. The substitution 'inner limit, then outer limit' fails here in two ways: $[\cos x] \to 0$ suggests $\lim\limits_{y \to 0} f(y) = \dfrac{1}{2}$, which is wrong because the inner function equals 0 exactly; and $[\cos 0] = 1$ suggests $f(1) = 1 - \cos 1$, which ignores that the limit uses $x \neq 0$.

Second term. $\sin x \to 0$ and $\sin x \neq 0$ for $0 < |x| < \pi$, so $f(\sin x) = \dfrac{1 - \cos(\sin x)}{\sin^2 x}$ near 0, and with $y = \sin x$ this tends to $\lim\limits_{y \to 0} \dfrac{1 - \cos y}{y^2} = \lim\limits_{y \to 0} \dfrac{2 \sin^2(y/2)}{y^2} = \dfrac{1}{2}$.

Hence the required limit is $1 + \dfrac{1}{2} = \dfrac{3}{2}$. (Option (A) uses $\dfrac{1}{2}$ for both terms; option (D) uses $f(1)$ for the first.)

Correct Option: (B)

Question 07

English

The value of $\lim\limits_{x\to 1}\dfrac{\sqrt{x+3}+\sqrt[3]{x+7}-4}{\sqrt[3]{9-x}-2}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

$\lim\limits_{x\to 1}\dfrac{\sqrt{x+3}+\sqrt[3]{x+7}-4}{\sqrt[3]{9-x}-2}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

At $x=1$: numerator $\sqrt{4}+\sqrt[3]{8}-4=0$ and denominator $\sqrt[3]{8}-2=0$, a $\dfrac{0}{0}$ form.

Write the numerator as $(\sqrt{x+3}-2)+(\sqrt[3]{x+7}-2)$, each part vanishing at $x=1$.

Rationalise: $\sqrt{x+3}-2=\dfrac{(x+3)-4}{\sqrt{x+3}+2}=\dfrac{x-1}{\sqrt{x+3}+2}$.

With $a^3-b^3=(a-b)(a^2+ab+b^2)$, $a=\sqrt[3]{x+7}$, $b=2$: $\sqrt[3]{x+7}-2=\dfrac{(x+7)-8}{(x+7)^{2/3}+2(x+7)^{1/3}+4}=\dfrac{x-1}{(x+7)^{2/3}+2(x+7)^{1/3}+4}$.

Similarly $\sqrt[3]{9-x}-2=\dfrac{(9-x)-8}{(9-x)^{2/3}+2(9-x)^{1/3}+4}=\dfrac{-(x-1)}{(9-x)^{2/3}+2(9-x)^{1/3}+4}$.

Cancelling $x-1$ ($x\neq 1$) and letting $x\to 1$: the numerator factor tends to $\dfrac{1}{4}+\dfrac{1}{4+4+4}=\dfrac{1}{4}+\dfrac{1}{12}=\dfrac{1}{3}$ and the denominator factor to $-\dfrac{1}{12}$.

Hence the limit is $\dfrac{1/3}{-1/12}=-4$.

Final Answer: -4

Question 08

English

For $x > 0$, let $Q(x) = \sqrt{\dfrac{x^2 + (x+8)^2}{2}}$ and $H(x) = \dfrac{2x(x+8)}{2x+8}$ denote the root mean square and the harmonic mean, respectively, of the numbers x and $x+8$. Then the value of $\lim\limits_{x\to\infty} x\big(Q(x) - H(x)\big)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

$x > 0$ के लिए माना कि $Q(x) = \sqrt{\dfrac{x^2 + (x+8)^2}{2}}$ तथा $H(x) = \dfrac{2x(x+8)}{2x+8}$ क्रमशः संख्याओं x तथा $x+8$ के वर्ग-माध्य-मूल तथा हरात्मक माध्य हैं। तब $\lim\limits_{x\to\infty} x\big(Q(x) - H(x)\big)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Both means tend to infinity, so $Q - H$ is an $\infty - \infty$ form; compare each with the arithmetic mean $A = \dfrac{x + (x+8)}{2} = x + 4$.

$Q^2 = \dfrac{x^2 + x^2 + 16x + 64}{2} = x^2 + 8x + 32$ and $A^2 = x^2 + 8x + 16$, so $Q^2 - A^2 = 16$.

Rationalising, $Q - A = \dfrac{Q^2 - A^2}{Q + A} = \dfrac{16}{Q + A}$, hence $x(Q - A) = \dfrac{16x}{\sqrt{x^2 + 8x + 32} + x + 4}$. Dividing numerator and denominator by x (the dominant term): $\dfrac{16}{\sqrt{1 + \frac{8}{x} + \frac{32}{x^2}} + 1 + \frac{4}{x}} \to \dfrac{16}{2} = 8$.

$H = \dfrac{2x(x+8)}{2x+8} = \dfrac{x(x+8)}{x+4}$, so $A - H = \dfrac{(x+4)^2 - x^2 - 8x}{x+4} = \dfrac{16}{x+4}$ and $x(A - H) = \dfrac{16x}{x+4} \to 16$.
Therefore $x(Q - H) = x(Q - A) + x(A - H) \to 8 + 16 = 24$.

(Check of size: for two numbers at distance d, each of $Q - A$, $A - G$, $G - H$ is about $\dfrac{d^2}{8x}$, giving $\dfrac{3d^2}{8} = 24$ for $d = 8$.)

Final Answer: 24

Question 09

English

Let $L = \lim\limits_{x\to 0}\left(\dfrac{2\log_e(1+x)}{e^{2x}-1}\right)^{1/x}$. Then the value of $\log_e L$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $L = \lim\limits_{x\to 0}\left(\dfrac{2\log_e(1+x)}{e^{2x}-1}\right)^{1/x}$ है। तब $\log_e L$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let $g(x) = \dfrac{2\ln(1+x)}{e^{2x}-1}$. Using $\dfrac{\ln(1+x)}{x} \to 1$ and $\dfrac{e^{2x}-1}{2x} \to 1$, $g(x) \to 1$, while $\dfrac{1}{x} \to \pm\infty$: the form is 1^{∞}.

For such a form, $L = e^{\lambda}$ with $\lambda = \lim\limits_{x\to 0}\dfrac{g(x)-1}{x}$, so $\ln L = \lambda$.

Expansions: $\ln(1+x) = x - \dfrac{x^2}{2} + O(x^3)$, so $2\ln(1+x) = 2x\left(1 - \dfrac{x}{2} + O(x^2)\right)$; $e^{2x} = 1 + 2x + 2x^2 + O(x^3)$, so $e^{2x} - 1 = 2x(1 + x + O(x^2))$.

Hence $g(x) = \dfrac{1 - \frac{x}{2} + O(x^2)}{1 + x + O(x^2)} = \left(1 - \dfrac{x}{2}\right)(1-x) + O(x^2) = 1 - \dfrac{3x}{2} + O(x^2)$.

So $\dfrac{g(x)-1}{x} \to -\dfrac{3}{2}$ from both sides, and $\ln L = -\dfrac{3}{2} = -1.50$.

Final Answer: -1.50

Question 10

English

For $n \in \mathbb{N}$, let $a_n = \dfrac{1}{n}\displaystyle\sum_{k=1}^{n} e^{k^3/n^4}$. If $\lim\limits_{n\to\infty} (a_n)^n = e^{\lambda}$, then the value of λ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

$n \in \mathbb{N}$ के लिए माना कि $a_n = \dfrac{1}{n}\displaystyle\sum_{k=1}^{n} e^{k^3/n^4}$ है। यदि $\lim\limits_{n\to\infty} (a_n)^n = e^{\lambda}$ है, तो λ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

The sum has no closed form, so a_n is squeezed. Let $t_k = \dfrac{k^3}{n^4}$; for $1 \le k \le n$, $0 < t_k \le \dfrac{n^3}{n^4} = \dfrac{1}{n}$.

Bounds on e^t for $0 \le t \le 1$: $e^t \ge 1 + t$, and $e^t = 1 + t + \dfrac{t^2}{2!} + \dfrac{t^3}{3!} + \cdots \le 1 + t + t^2\left(\dfrac{1}{2} + \dfrac{1}{6} + \ldots\right) \le 1 + t + t^2$ (since $e - 2 < 1$).

Sum of cubes: $\displaystyle\sum_{k=1}^{n} k^3 = \frac{n^2(n+1)^2}{4}$, so $\sum t_k = \dfrac{(n+1)^2}{4n^2}$. Also $\sum t_k^2 \le n \cdot \dfrac{1}{n^2} = \dfrac{1}{n}$.

Dividing by n: $1 + \dfrac{(n+1)^2}{4n^3} \le a_n \le 1 + \dfrac{(n+1)^2}{4n^3} + \dfrac{1}{n^2}$. Both bounds tend to 1, so $a_n \to 1$ and $(a_n)^n$ is a 1^∞ form.

For a 1^∞ form, $\lim (a_n)^n = e^{\lim n(a_n - 1)}$. Multiplying the bounds by n: $\dfrac{(n+1)^2}{4n^2} \le n(a_n - 1) \le \dfrac{(n+1)^2}{4n^2} + \dfrac{1}{n}$.

Both sides tend to $\dfrac{1}{4}$, so by the sandwich theorem $n(a_n - 1) \to \dfrac{1}{4}$.

Hence $\lim (a_n)^n = e^{1/4}$ and $\lambda = 0.25$.

Final Answer: 0.25

Question 11

English

For a real number t, let $\{t\} = t - [t]$, where $[t]$ denotes the greatest integer less than or equal to t. Then the value of $\lim\limits_{n\to\infty} n\left(1 - \left\{\sqrt{n^2+40n}\right\}\right)$, where n takes natural number values, is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

वास्तविक संख्या t के लिए माना कि $\{t\} = t - [t]$ है, जहाँ $[t]$, t से कम या उसके बराबर महत्तम पूर्णांक को दर्शाता है। तब $\lim\limits_{n\to\infty} n\left(1 - \left\{\sqrt{n^2+40n}\right\}\right)$ का मान, जहाँ n प्राकृत संख्या मान लेता है, _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Since $n^2 + 40n = (n+20)^2 - 400 < (n+20)^2$, we have $\sqrt{n^2+40n} < n + 20$.

Also $(n+19)^2 = n^2 + 38n + 361 < n^2 + 40n$ exactly when $2n > 361$, i.e. $n \geq 181$. So for $n \geq 181$, $n + 19 < \sqrt{n^2+40n} < n + 20$ and $[\sqrt{n^2+40n}] = n + 19$.

Then $\{\sqrt{n^2+40n}\} = \sqrt{n^2+40n} - (n+19)$, and $1 - \{\sqrt{n^2+40n}\} = n + 20 - \sqrt{n^2+40n}$.

Rationalising: $n + 20 - \sqrt{n^2+40n} = \dfrac{(n+20)^2 - (n^2+40n)}{n+20+\sqrt{n^2+40n}} = \dfrac{400}{n+20+\sqrt{n^2+40n}}$.

Hence $n\left(1 - \{\sqrt{n^2+40n}\}\right) = \dfrac{400}{1 + \frac{20}{n} + \sqrt{1+\frac{40}{n}}} \to \dfrac{400}{2} = 200$.

Note the trap: $\sqrt{n^2+40n} \approx n + 20$ suggests a fractional part near 0, but the root is always just below $n + 20$, so the fractional part tends to 1.

Final Answer: 200

Question 12

English

Let a and b be real numbers and let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = [x](x - a) + [2x](x - b)$, where $[t]$ denotes the greatest integer less than or equal to t. If f is continuous at every point of the interval $\left(\frac{5}{4}, \frac{9}{4}\right)$, then the value of $\lim\limits_{x \to 3^-} f(x)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि a तथा b वास्तविक संख्याएँ हैं तथा $f : \mathbb{R} \to \mathbb{R}$, $f(x) = [x](x - a) + [2x](x - b)$ द्वारा परिभाषित है, जहाँ $[t]$, t से कम या उसके बराबर महत्तम पूर्णांक को दर्शाता है। यदि f अन्तराल $\left(\frac{5}{4}, \frac{9}{4}\right)$ के प्रत्येक बिन्दु पर सतत है, तो $\lim\limits_{x \to 3^-} f(x)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

f is continuous wherever both $[x]$ and $[2x]$ are locally constant. In $\left(\frac{5}{4}, \frac{9}{4}\right)$, $[x]$ changes only at $x = 2$, and $[2x]$ changes only where $2x$ is an integer, i.e. at $x = \frac{3}{2}$ and $x = 2$.

At $x = \frac{3}{2}$: $[x] = 1$ near $\frac{3}{2}$; $[2x] = 2$ for $x < \frac{3}{2}$ and 3 for $x \geq \frac{3}{2}$. So $\lim\limits_{x \to 3/2^-} f = \frac{1}{2} - a + 2\left(\frac{3}{2} - b\right)$ and $f\left(\frac{3}{2}\right) = \frac{1}{2} - a + 3\left(\frac{3}{2} - b\right)$. Continuity gives $\frac{3}{2} - b = 0$, so $b = \frac{3}{2}$.

At $x = 2$: for $x < 2$, $[x] = 1$ and $[2x] = 3$; for $x \geq 2$, $[x] = 2$ and $[2x] = 4$. So $\lim\limits_{x \to 2^-} f = (2 - a) + 3(2 - b)$ and $f(2) = 2(2 - a) + 4(2 - b)$. Continuity gives $(2 - a) + (2 - b) = 0$, so $a = 4 - b = \frac{5}{2}$.

With these values f is continuous on the whole interval (the end points $\frac{5}{4}$, $\frac{9}{4}$ are not jump points, and $x = 1$, $x = \frac{5}{2}$ lie outside).

Left-hand limit at 3: for $\frac{5}{2} \leq x < 3$, $[x] = 2$ and $[2x] = 5$, so $f(x) = 2\left(x - \frac{5}{2}\right) + 5\left(x - \frac{3}{2}\right) \to 2 \cdot \frac{1}{2} + 5 \cdot \frac{3}{2} = 1 + \frac{15}{2} = \frac{17}{2}$.

Hence $\lim\limits_{x \to 3^-} f(x) = 8.50$ (whereas $f(3) = 3 \cdot \frac{1}{2} + 6 \cdot \frac{3}{2} = 10.5$).

Final Answer: 8.50

Question 13

English

Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(t) = 4t^3 - 15t^2 + 12t$, and let $g : [0, \infty) \to \mathbb{R}$ be defined by $g(x) = \max\{f(t) : 0 \le t \le x\}$. Let c be the point of $(0, \infty)$ at which g is not differentiable. Then the value of $\lim\limits_{h \to 0^+} \dfrac{g(c+h) - g(c-h)}{h}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $f : \mathbb{R} \to \mathbb{R}$, $f(t) = 4t^3 - 15t^2 + 12t$ द्वारा परिभाषित है तथा $g : [0, \infty) \to \mathbb{R}$, $g(x) = \max\{f(t) : 0 \le t \le x\}$ द्वारा परिभाषित है। माना कि c, $(0, \infty)$ का वह बिन्दु है जिस पर g अवकलनीय नहीं है। तब $\lim\limits_{h \to 0^+} \dfrac{g(c+h) - g(c-h)}{h}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$f'(t) = 12t^2 - 30t + 12 = 6(2t^2 - 5t + 2) = 6(2t-1)(t-2)$. So f increases on $\left[0, \dfrac{1}{2}\right]$, decreases on $\left[\dfrac{1}{2}, 2\right]$ and increases on $[2, \infty)$, with $f(0) = 0$, $f\left(\dfrac{1}{2}\right) = \dfrac{1}{2} - \dfrac{15}{4} + 6 = \dfrac{11}{4}$ and $f(2) = 32 - 60 + 24 = -4$.

The running maximum $g(x)$ equals $f(x)$ while f is increasing from 0, i.e. on $\left[0, \dfrac{1}{2}\right]$. After that it stays at $\dfrac{11}{4}$ until f climbs back to $\dfrac{11}{4}$.

$f(t) - \dfrac{11}{4} = 4t^3 - 15t^2 + 12t - \dfrac{11}{4}$ has the double root $t = \dfrac{1}{2}$ (local maximum), and dividing gives $\left(t - \dfrac{1}{2}\right)^2 (4t - 11)$. So f regains the value $\dfrac{11}{4}$ at $t = \dfrac{11}{4}$, and $g = f$ on $\left[\dfrac{11}{4}, \infty\right)$ since f is increasing there.

Hence $g(x) = f(x)$ for $0 \le x \le \dfrac{1}{2}$, $g(x) = \dfrac{11}{4}$ for $\dfrac{1}{2} \le x \le \dfrac{11}{4}$, and $g(x) = f(x)$ for $x \ge \dfrac{11}{4}$.

Differentiability: at $x = \dfrac{1}{2}$ both one-sided derivatives are 0 (as $f'\left(\dfrac{1}{2}\right) = 0$), so g is differentiable there. At $x = \dfrac{11}{4}$ the left derivative is 0 and the right derivative is $f'\left(\dfrac{11}{4}\right) = 6\left(\dfrac{11}{2} - 1\right)\left(\dfrac{11}{4} - 2\right) = 6 \cdot \dfrac{9}{2} \cdot \dfrac{3}{4} = \dfrac{81}{4}$. So $c = \dfrac{11}{4}$ is the only point of non-differentiability.

Write $\dfrac{g(c+h) - g(c-h)}{h} = \dfrac{g(c+h) - g(c)}{h} + \dfrac{g(c) - g(c-h)}{h}$. As $h \to 0^+$ the first term tends to the right derivative $\dfrac{81}{4}$ and the second to the left derivative 0.

Hence the limit is $\dfrac{81}{4} = 20.25$.

Final Answer: 20.25

Question 14

English

Let $f : (0, \infty) \to \mathbb{R}$ be defined by $f(x) = \lim\limits_{n\to\infty} \dfrac{x^n \log_e x + \sqrt{x} - 1}{x^n + 1}$. Then the value of $\lim\limits_{h\to 0} \dfrac{f(1+h)\, f(1-h)}{h^2}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $f : (0, \infty) \to \mathbb{R}$, $f(x) = \lim\limits_{n\to\infty} \dfrac{x^n \log_e x + \sqrt{x} - 1}{x^n + 1}$ द्वारा परिभाषित है। तब $\lim\limits_{h\to 0} \dfrac{f(1+h)\, f(1-h)}{h^2}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Finding f. For $x > 1$, divide numerator and denominator by x^n: $f(x) = \lim \dfrac{\ln x + (\sqrt{x} - 1)x^{-n}}{1 + x^{-n}} = \ln x$. For $0 < x < 1$, $x^n \to 0$, so $f(x) = \sqrt{x} - 1$. At $x = 1$, $f(1) = \dfrac{0+0}{1+1} = 0$.

So $f(x) = \sqrt{x} - 1$ on $(0, 1)$, $f(1) = 0$, $f(x) = \ln x$ on $(1, \infty)$: a function given by different formulas on the two sides of 1.

Right side, $0 < h < 1$: $1 + h > 1$ and $1 - h < 1$, so $\dfrac{f(1+h)f(1-h)}{h^2} = \dfrac{\ln(1+h)}{h} \cdot \dfrac{\sqrt{1-h} - 1}{h}$.

Here $\dfrac{\ln(1+h)}{h} \to 1$ and $\dfrac{\sqrt{1-h} - 1}{h} = \dfrac{-1}{\sqrt{1-h} + 1} \to -\dfrac{1}{2}$; the product tends to $-\dfrac{1}{2}$.

Left side, $-1 < h < 0$: now $1 + h < 1$ and $1 - h > 1$, so the quotient is $\dfrac{\sqrt{1+h} - 1}{h} \cdot \dfrac{\ln(1-h)}{h}$. Here $\dfrac{\sqrt{1+h} - 1}{h} = \dfrac{1}{\sqrt{1+h} + 1} \to \dfrac{1}{2}$ and $\dfrac{\ln(1-h)}{h} \to -1$; the product tends to $-\dfrac{1}{2}$.

The one-sided limits agree although f itself has different one-sided slopes at 1 (1 and $\dfrac{1}{2}$). Hence the limit is $-\dfrac{1}{2} = -0.50$.

Final Answer: -0.50

Question 15

English

Let a, b and c be real numbers with $c > 0$, and let $f : \mathbb{R} \to \mathbb{R}$ be defined by

$$f(x) = \begin{cases} x^4 - 8x^2 + 4x, & |x| \geq c \\ ax + b, & |x| < c \end{cases}$$

If f is differentiable at every point of $\mathbb{R}$, then the value of $f(3) - f(1)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि a, b तथा c वास्तविक संख्याएँ हैं जहाँ $c > 0$ है, तथा $f : \mathbb{R} \to \mathbb{R}$ निम्न द्वारा परिभाषित है

$$f(x) = \begin{cases} x^4 - 8x^2 + 4x, & |x| \geq c \\ ax + b, & |x| < c \end{cases}$$

यदि f, $\mathbb{R}$ के प्रत्येक बिन्दु पर अवकलनीय है, तो $f(3) - f(1)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let $q(x) = x^4 - 8x^2 + 4x$, so $q'(x) = 4x^3 - 16x + 4$. Inside $(-c, c)$, f is the line $ax + b$; outside, $f = q$. Both pieces are differentiable, so only the junctions $x = c$ and $x = -c$ matter.

At $x = c$: continuity needs $\lim_{x \to c^-} f(x) = ac + b = q(c)$, and equal one-sided derivatives need $a = q'(c)$.

At $x = -c$: $-ac + b = q(-c)$ and $a = q'(-c)$.

From the slopes, $q'(c) = q'(-c)$: $4c^3 - 16c + 4 = -4c^3 + 16c + 4$, so $8c^3 - 32c = 0$, $8c(c^2 - 4) = 0$. With $c > 0$, $c = 2$.

Then $a = q'(2) = 32 - 32 + 4 = 4$. From $x = 2$: $b = q(2) - 2a = (16 - 32 + 8) - 8 = -16$. Check at $x = -2$: $q(-2) = 16 - 32 - 8 = -24$ and $-2a + b = -8 - 16 = -24$, consistent. So $a = 4$, $b = -16$, $c = 2$ is the unique choice.

Geometrically, the line $y = 4x - 16$ is tangent to $y = q(x)$ at both $x = -2$ and $x = 2$: indeed $q(x) - (4x - 16) = (x^2 - 4)^2$.

Since $|1| < 2$, $f(1) = a + b = 4 - 16 = -12$; since $|3| \geq 2$, $f(3) = q(3) = 81 - 72 + 12 = 21$.

Therefore $f(3) - f(1) = 21 - (-12) = 33$.

Final Answer: 33

Question 16

English

Let $f : \mathbb{R} \setminus \{1\} \to \mathbb{R}$ be defined by $f(x) = \dfrac{x^{20} - 1}{x - 1}$. Then the value of $\lim_{x \to 0} \dfrac{f(f(x)) - 20}{x}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $f : \mathbb{R} \setminus \{1\} \to \mathbb{R}$, $f(x) = \dfrac{x^{20} - 1}{x - 1}$ द्वारा परिभाषित है। तब $\lim_{x \to 0} \dfrac{f(f(x)) - 20}{x}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

For $x \neq 1$, $f(x) = 1 + x + x^2 + \cdots + x^{19}$. Thus $f(0) = 1$, which is outside the domain of f, so $f \circ f$ is not defined at $x = 0$; the limit uses only $x \neq 0$ near 0.

Put $u = f(x) - 1 = x + x^2 + \cdots + x^{19} = x(1 + x + \cdots + x^{18})$. As $x \to 0$, $u \to 0$, and for $0 < |x| < \dfrac{1}{2}$ the bracket is positive, so $u \neq 0$; hence $f(x) \neq 1$ and $f(f(x))$ is defined.

Near $y = 1$ (with $y \neq 1$), $f(y) = \sum_{k=0}^{19} y^k$. With $y = 1 + u$, the binomial expansion gives $(1 + u)^k = 1 + ku + O(u^2)$, so $f(1 + u) = 20 + u(1 + 2 + \cdots + 19) + O(u^2) = 20 + 190u + O(u^2)$.

So $\dfrac{f(f(x)) - 20}{x} = 190 \cdot \dfrac{u}{x} + \dfrac{O(u^2)}{x}$. Since $\dfrac{u}{x} = 1 + x + \cdots + x^{18} \to 1$ and $\dfrac{u^2}{x} = u \cdot \dfrac{u}{x} \to 0$, the limit is 190.

(Substituting first, $f(f(0)) = f(1)$, is impossible; the limit exists because $f(x) \to 1$ without ever equalling 1.)

Final Answer: 190

Question 17

English

Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = x^2 - 2x$ and let $g : \mathbb{R} \setminus \{-1\} \to \mathbb{R}$ be defined by $g(y) = \dfrac{y^6 - 1}{|y + 1|}$. Then the value of $\lim\limits_{x \to 1} g\big(f(x)\big)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2 - 2x$ द्वारा तथा $g : \mathbb{R} \setminus \{-1\} \to \mathbb{R}$, $g(y) = \dfrac{y^6 - 1}{|y + 1|}$ द्वारा परिभाषित हैं। तब $\lim\limits_{x \to 1} g\big(f(x)\big)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Factor: $y^6 - 1 = (y^2 - 1)(y^4 + y^2 + 1) = (y - 1)(y + 1)(y^4 + y^2 + 1)$. So for $y \neq -1$, $g(y) = (y - 1)(y^4 + y^2 + 1) \cdot \dfrac{y + 1}{|y + 1|}$.

The last factor is 1 for $y > -1$ and -1 for $y < -1$. Hence $\lim\limits_{y \to -1^+} g(y) = (-2)(3) = -6$ and $\lim\limits_{y \to -1^-} g(y) = 6$: g has no limit at -1, and $g(-1)$ is not defined.

The tempting substitution 'inner limit -1, then the limit of g at -1' therefore suggests that the limit does not exist. It must be refined by the side from which $f(x)$ approaches -1.

$f(x) + 1 = x^2 - 2x + 1 = (x - 1)^2 > 0$ for $x \neq 1$. So as $x \to 1$ (from either side), $f(x) \to -1$ through values greater than -1, and never equals -1.

Hence $\lim\limits_{x \to 1} g(f(x)) = \lim\limits_{y \to -1^+} g(y) = -6$. Directly: for $x \neq 1$, $g(f(x)) = (f(x) - 1)\big(f(x)^4 + f(x)^2 + 1\big) \to (-2)(3) = -6$.

Final Answer: -6

Question 18

English

The value of $\lim\limits_{x\to 0} \dfrac{\sin x\ \sin^{-1} x - x^2}{\tan x\ \tan^{-1} x - x^2}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

$\lim\limits_{x\to 0} \dfrac{\sin x\ \sin^{-1} x - x^2}{\tan x\ \tan^{-1} x - x^2}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Both numerator and denominator are $x^2 + (\text{higher terms}) - x^2$. In each product the x^3 terms of the two factors are opposite, so the x^4 terms cancel too and the first surviving term is of order x^6. L'Hopital's rule would need six differentiations of products with inverse functions, so Maclaurin series are used.

Series to order 5: $\sin x = x - \dfrac{x^3}{6} + \dfrac{x^5}{120} + \ldots$, $\sin^{-1} x = x + \dfrac{x^3}{6} + \dfrac{3x^5}{40} + \ldots$, $\tan x = x + \dfrac{x^3}{3} + \dfrac{2x^5}{15} + \ldots$, $\tan^{-1} x = x - \dfrac{x^3}{3} + \dfrac{x^5}{5} - \ldots$

Numerator: $\sin x\ \sin^{-1} x = x^2 + x^4\left(\dfrac{1}{6} - \dfrac{1}{6}\right) + x^6\left(\dfrac{3}{40} - \dfrac{1}{36} + \dfrac{1}{120}\right) + O(x^8)$. With denominator 360: $\dfrac{27 - 10 + 3}{360} = \dfrac{20}{360} = \dfrac{1}{18}$. So the numerator is $\dfrac{x^6}{18} + O(x^8)$.

Denominator: $\tan x\ \tan^{-1} x = x^2 + x^4\left(\dfrac{1}{3} - \dfrac{1}{3}\right) + x^6\left(\dfrac{1}{5} - \dfrac{1}{9} + \dfrac{2}{15}\right) + O(x^8)$. With denominator 45: $\dfrac{9 - 5 + 6}{45} = \dfrac{10}{45} = \dfrac{2}{9}$. So the denominator is $\dfrac{2x^6}{9} + O(x^8)$.

Therefore the limit is $\dfrac{1/18}{2/9} = \dfrac{9}{36} = \dfrac{1}{4} = 0.25$.

Final Answer: 0.25

Question 19

English

Let $f : \mathbb{R} \to \mathbb{R}$ be a function that is continuous at $x = 0$, with $f(0) = -1$ and $f(x) - f\left(\frac{x}{2}\right) = \frac{3x^2}{4}$ for all $x \in \mathbb{R}$. Let $g : \mathbb{R} \to \mathbb{R}$ be defined by $g(x) = \lim_{n\to\infty} \left(1 + |f(x)|^n\right)^{1/n}$. Then the product of all real numbers at which g is not differentiable is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $f : \mathbb{R} \to \mathbb{R}$ एक फलन है जो $x = 0$ पर सतत है, $f(0) = -1$ है तथा सभी $x \in \mathbb{R}$ के लिए $f(x) - f\left(\frac{x}{2}\right) = \frac{3x^2}{4}$ है। माना कि $g : \mathbb{R} \to \mathbb{R}$, $g(x) = \lim_{n\to\infty} \left(1 + |f(x)|^n\right)^{1/n}$ द्वारा परिभाषित है। तब उन सभी वास्तविक संख्याओं का गुणनफल, जिन पर g अवकलनीय नहीं है, _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Finding f. Applying the relation at $\frac{x}{2^k}$: $f\left(\frac{x}{2^k}\right) - f\left(\frac{x}{2^{k+1}}\right) = \frac{3x^2}{4 \cdot 4^k}$. Adding for $k = 0, 1, \ldots, m-1$ telescopes to $f(x) - f\left(\frac{x}{2^m}\right) = \frac{3x^2}{4} \cdot \frac{1 - 4^{-m}}{1 - \frac{1}{4}} = x^2\left(1 - 4^{-m}\right)$.

As $m \to \infty$, $\frac{x}{2^m} \to 0$ and, because f is continuous at 0, $f\left(\frac{x}{2^m}\right) \to f(0) = -1$. Hence $f(x) = x^2 - 1$ for every x. (Without continuity at 0 the relation alone would not determine f.)

Finding g. For $t = |f(x)| \geq 0$: if $t < 1$, $(1 + t^n)^{1/n} \to 1$; if $t = 1$, $2^{1/n} \to 1$; if $t > 1$, $(1 + t^n)^{1/n} = t(1 + t^{-n})^{1/n} \to t$. So $g(x) = \max\{1, |x^2 - 1|\}$.

$|x^2 - 1| > 1$ means $x^2 > 2$ (the other case $x^2 < 0$ is impossible). So $g(x) = x^2 - 1$ for $|x| > \sqrt{2}$ and $g(x) = 1$ for $|x| \leq \sqrt{2}$.

At $x = \pm\sqrt{2}$, g is continuous (value 1) but the one-sided derivatives are 0 and $2x = \pm 2\sqrt{2}$, so g is not differentiable there. At $x = 0$, $|f(0)| = 1$, but $|x^2 - 1| = 1 - x^2 \leq 1$ nearby, so $g \equiv 1$ near 0 and is differentiable. Elsewhere g is a polynomial or constant.

Points of non-differentiability: $\sqrt{2}$ and $-\sqrt{2}$; their product is -2.

Final Answer: -2

Question 20

English

Let a and b be real numbers such that $\lim_{x\to 0} \dfrac{(1+ax)e^x - 1 - bx}{x^3}$ exists and is a finite real number. Then the value of ab is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि a तथा b ऐसी वास्तविक संख्याएँ हैं कि $\lim_{x\to 0} \dfrac{(1+ax)e^x - 1 - bx}{x^3}$ का अस्तित्व है तथा यह एक परिमित वास्तविक संख्या है। तब ab का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Expand $e^x = 1 + x + \dfrac{x^2}{2} + \dfrac{x^3}{6} + O(x^4)$. Then $(1+ax)e^x = 1 + (1+a)x + \left(\dfrac{1}{2} + a\right)x^2 + \left(\dfrac{1}{6} + \dfrac{a}{2}\right)x^3 + O(x^4)$.

So the numerator is $N(x) = (1 + a - b)x + \left(\dfrac{1}{2} + a\right)x^2 + \left(\dfrac{1}{6} + \dfrac{a}{2}\right)x^3 + O(x^4)$.

$\dfrac{N(x)}{x^3} = \dfrac{1+a-b}{x^2} + \dfrac{\frac{1}{2}+a}{x} + \left(\dfrac{1}{6} + \dfrac{a}{2}\right) + O(x)$. If $1 + a - b \neq 0$ the quotient is unbounded; if $1 + a - b = 0$ but $\dfrac{1}{2} + a \neq 0$, it is still unbounded. So both coefficients must vanish.

$\dfrac{1}{2} + a = 0 \Rightarrow a = -\dfrac{1}{2}$, and $b = 1 + a = \dfrac{1}{2}$. Then the limit equals $\dfrac{1}{6} - \dfrac{1}{4} = -\dfrac{1}{12}$, which is finite, so these values work and are the only ones.

Hence $ab = -\dfrac{1}{2} \cdot \dfrac{1}{2} = -\dfrac{1}{4} = -0.25$.

Final Answer: -0.25

Question 21

English

Let a and b be positive real numbers such that $\lim\limits_{x\to 0}\dfrac{\sqrt[3]{1+ax}-\sqrt{1+bx}}{x^2}=-18$. Then the value of ab is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि a तथा b ऐसी धनात्मक वास्तविक संख्याएँ हैं कि $\lim\limits_{x\to 0}\dfrac{\sqrt[3]{1+ax}-\sqrt{1+bx}}{x^2}=-18$ है। तब ab का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Binomial expansions for small x: $(1+ax)^{1/3}=1+\dfrac{1}{3}ax+\dfrac{\frac{1}{3}\left(-\frac{2}{3}\right)}{2}a^2x^2+O(x^3)=1+\dfrac{ax}{3}-\dfrac{a^2x^2}{9}+O(x^3)$

and $(1+bx)^{1/2}=1+\dfrac{bx}{2}-\dfrac{b^2x^2}{8}+O(x^3)$.

The numerator is $\left(\dfrac{a}{3}-\dfrac{b}{2}\right)x+\left(\dfrac{b^2}{8}-\dfrac{a^2}{9}\right)x^2+O(x^3)$.

Divided by x^2, the term $\left(\dfrac{a}{3}-\dfrac{b}{2}\right)\dfrac{1}{x}$ is unbounded unless $\dfrac{a}{3}=\dfrac{b}{2}$. So $a=\dfrac{3b}{2}$.

Then the limit is $\dfrac{b^2}{8}-\dfrac{a^2}{9}=\dfrac{b^2}{8}-\dfrac{9b^2/4}{9}=\dfrac{b^2}{8}-\dfrac{b^2}{4}=-\dfrac{b^2}{8}$.

$-\dfrac{b^2}{8}=-18\Rightarrow b^2=144\Rightarrow b=12$ (as $b>0$), and $a=\dfrac{3}{2}\cdot 12=18$.

Hence $ab=18\times 12=216$.

Final Answer: 216

Question 22

English

For each integer $n \geq 2$, let x_n be the unique positive real number satisfying $x_n + x_n^2 + x_n^3 + \cdots + x_n^n = \frac{3}{2}$. Then the value of $\lim_{n\to\infty} x_n$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

प्रत्येक पूर्णांक $n \geq 2$ के लिए माना कि x_n वह अद्वितीय धनात्मक वास्तविक संख्या है जो $x_n + x_n^2 + x_n^3 + \cdots + x_n^n = \frac{3}{2}$ को संतुष्ट करती है। तब $\lim_{n\to\infty} x_n$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let $S_n(x) = x + x^2 + \cdots + x^n$. It is strictly increasing for $x > 0$, with $S_n(0) = 0$ and $S_n(1) = n \geq 2 > \frac{3}{2}$, so the root x_n exists, is unique and lies in $(0, 1)$.

For $0 < x < 1$, $S_n(x) = \frac{x(1-x^n)}{1-x}$. Setting this equal to $\frac{3}{2}$: $2x - 2x^{n+1} = 3 - 3x$, i.e. $x_n = \frac{3}{5} + \frac{2}{5}x_n^{n+1}$.

Bound: since $S_n(x) \geq x + x^2$, we get $x_n + x_n^2 \leq \frac{3}{2}$, so $x_n \leq x_2$, where $x_2^2 + x_2 - \frac{3}{2} = 0$ gives $x_2 = \frac{-1+\sqrt{7}}{2} \approx 0.82$.

Hence $0 < x_n^{n+1} \leq x_2^{n+1} \to 0$ as $n \to \infty$ (a fixed base less than 1).

Therefore $x_n = \frac{3}{5} + \frac{2}{5}x_n^{n+1} \to \frac{3}{5} = 0.60$.

(Treating the equation as the infinite series $\frac{x}{1-x} = \frac{3}{2}$ gives the right value, but the uniform bound $x_n \leq x_2 < 1$ is what justifies $x_n^{n+1} \to 0$.)

Final Answer: 0.60

Question 23

English

The value of $\lim\limits_{n\to\infty}\left(\sum\limits_{k=1}^{n}\left(1+\dfrac{k}{2n}\right)^{2n}\right)^{1/n}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

$\lim\limits_{n\to\infty}\left(\sum\limits_{k=1}^{n}\left(1+\dfrac{k}{2n}\right)^{2n}\right)^{1/n}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

For $1 \le k \le n$, $1+\dfrac{k}{2n}$ increases with k, so every term is at most the last one, $\left(1+\dfrac{n}{2n}\right)^{2n} = \left(\dfrac{3}{2}\right)^{2n} = \left(\dfrac{9}{4}\right)^{n}$, and the sum is at least that last term.

Hence $\left(\dfrac{9}{4}\right)^{n} \le \sum\limits_{k=1}^{n}\left(1+\dfrac{k}{2n}\right)^{2n} \le n\left(\dfrac{9}{4}\right)^{n}$.

Taking n-th roots (an increasing operation): $\dfrac{9}{4} \le \left(\sum\limits_{k=1}^{n}\left(1+\dfrac{k}{2n}\right)^{2n}\right)^{1/n} \le \dfrac{9}{4}\,n^{1/n}$.

$n^{1/n} = e^{\frac{\ln n}{n}} \to e^{0} = 1$, so the right bound tends to $\dfrac{9}{4}$.

By the sandwich theorem the limit is $\dfrac{9}{4} = 2.25$. (Replacing each term by its own limit e^{k} is wrong: the terms depend on n and the sum is governed by its largest term.)

Final Answer: 2.25